

BIST Factor Investing Project

- Group participants: [set project.group_participants in config/project_config.yaml].
- Project focus: Turkish equities, BIST100 benchmark, ROE, P/E, momentum, and trend factors.
- Data-source run: Yahoo Finance frozen CSV.
- All data and results are frozen to CSV for reproducibility.

Paper Summary

- Han, Zhou, and Zhu ask whether combining price information across short, intermediate, and long horizons improves factor returns.
- Their trend factor uses moving averages from 3 to 1,000 trading days, predicts expected returns, buys the highest predicted-return quintile, and shorts the lowest.
- They report stronger Sharpe ratios than separate reversal, momentum, and long-term reversal factors, including resilience during the financial crisis.
- Our course adaptation keeps the moving-average logic but estimates the predictive regression on BIST100 Index data as required by the project PDF.

Data and Factor Construction

- Use XU100 only for benchmark and trend regression; stock portfolios exclude index columns.
- ROE and P/E come from the frozen fundamental or BIST ratio file and are lagged by using month t signals for month $t+1$.
- Momentum is $P_t / P_{t-12} - 1$ using month-end closes from the frozen price file.
- All factor signals are winsorized and cross-sectionally z-scored before portfolio formation.

Trend Regression

- Trend uses only XU100/BIST100 for coefficient estimation; index columns are excluded from stock portfolios.
- The configured primary regression excludes the common intercept because z-scoring/ranking removes any stock-common constant.
- Selected mad_1000: coef 0.0240, p-value 0.0008.
- With-intercept diagnostics are saved separately to disclose specification sensitivity.

Code Example: Factors

- All factors are constructed at month end.
- Signals at month t are matched to returns/trades at month $t+1$.

```
def compute_momentum(monthly_close, stock_cols):  
    return monthly_close[stock_cols] / monthly_close[stock_cols].shift(12) - 1  
  
def zscore_cross_section(s):  
    s = winsorize_cross_section(s, lower_q=0.01, upper_q=0.99)  
    valid = s.dropna()  
    if len(valid) < 3 or valid.std(ddof=1) == 0:  
        return pd.Series(np.nan, index=s.index)  
    return (s - valid.mean()) / valid.std(ddof=1)  
  
target = monthly_returns["XU100"].shift(-1)  
model = sm.OLS(target, xu100_ma_deviations[selected_terms]).fit()
```

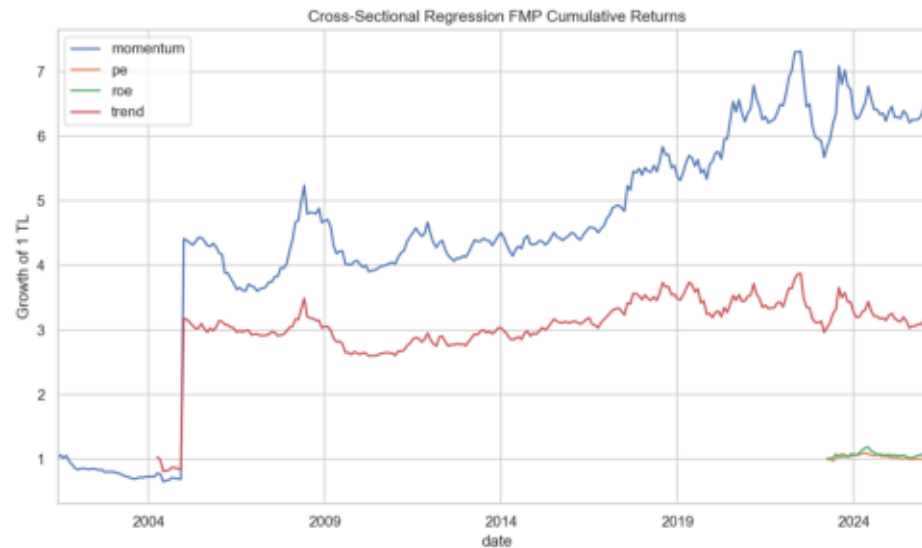
Factor-Mimicking Portfolios

- Each factor is analyzed with both portfolio-sort FMPs and cross-sectional-regression FMPs.
- Best factor by Sharpe in the saved metrics: pe_portfolio_sort.
- Saved outputs include separate-start metrics, common-start metrics, and selected-date portfolio/regression weights.
- Method comparison reports return correlations and same-sign frequencies between portfolio and regression FMPs.



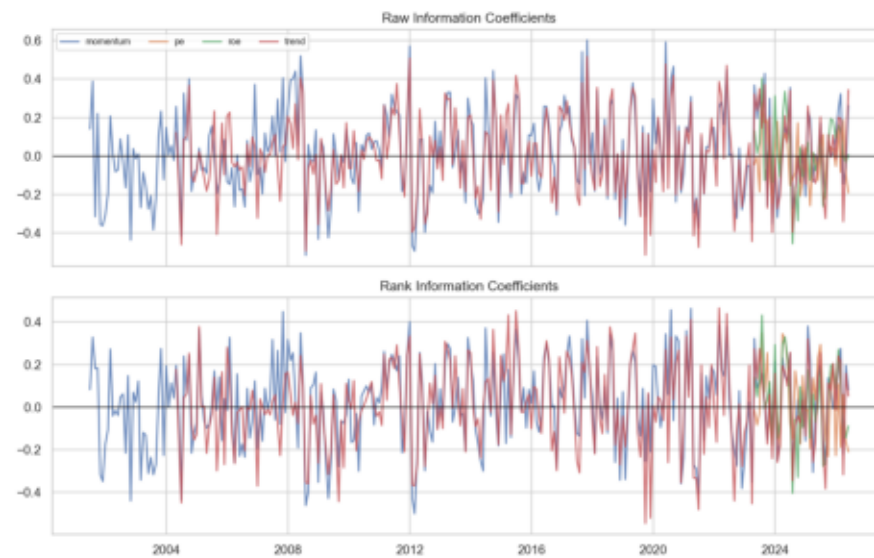
FMP Comparison Results

- pe / portfolio_sort: common-start Sharpe 0.62, avg monthly return 1.10%.
- momentum / cross_sectional_regression: common-start Sharpe 0.53, avg monthly return 0.46%.
- momentum / portfolio_sort: common-start Sharpe 0.48, avg monthly return 0.94%.
- Portfolio-sort and regression methods were qualitatively similar for 4 of 4 factors by the saved rule.



Information Coefficients

- Raw IC measures cross-sectional correlation between factor signal at month t and stock returns at month $t+1$.
- Rank IC repeats the test on ranks and is more robust to outliers.
- Positive average IC suggests the factor ranks stocks in the correct direction.



Code Example: Backtrader

- This matches the instructor's capital, sizing, commission, and market-order rules.
- Trades, orders, positions, daily returns, and equity curves are saved to CSV.

```
class FixedCashSizer(bt.Sizer):
    params = (("cash_per_trade", 100_000),)
    def _getsizing(self, comminfo, cash, data, isbuy):
        price = data.close[0]
        return int(self.p.cash_per_trade / price) if price > 0 else 0

cerebro.broker.setcash(1_000_000)
cerebro.broker.setcommission(commission=0)
cerebro.addsizer(FixedCashSizer, cash_per_trade=100_000)
self.buy(data=data, exectype=bt.Order.Market)
```

Backtrader Strategy

- Monthly strategy buys the top-ranked composite-factor stocks.
- Initial capital is 1,000,000 TL; each trade uses the FixedCashSizer with 100,000 TL per trade.
- Commission is set to zero and base orders are market orders, matching the instructor instructions.
- Trades, positions, daily returns, equity curves, and summary metrics are saved as CSV files.



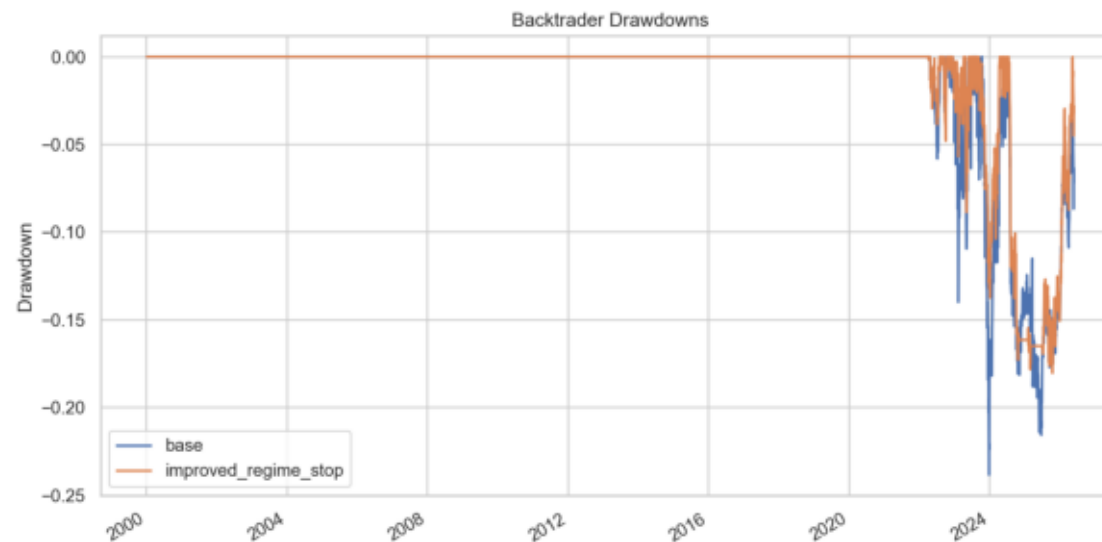
What We Tried

- Test grid: equal all four factors, price/trend, trend-heavy, and quality/value composites.
- For each composite, top 5 / top 10 / top 15 and BIST100 regime filter on/off were tested.
- Backtrader then tested the base strategy and a regime plus stop-loss/take-profit risk-control variant.
- Best vectorized trial: equal_all_top15_no_regime with Sharpe 1.47 and annualized return 54.26%.
- Walk-forward: 44 windows, compounded test return 13119.90%.



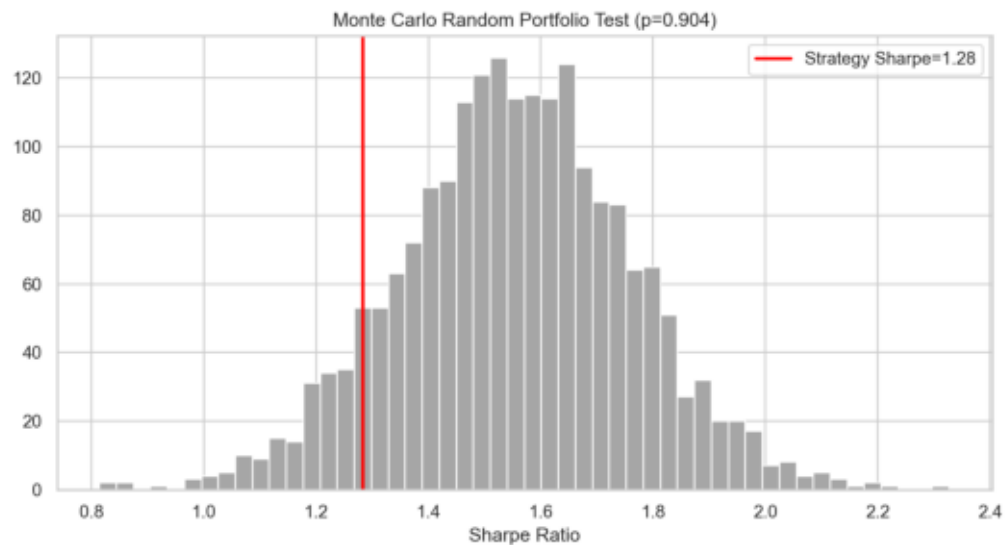
What Failed or Weakened

- The trend signal is specification-sensitive, so we disclose both the primary no-intercept selection and with-intercept diagnostic.
- Regime filtering and stop/take-profit rules reduced final wealth in the Backtrader run, even if they slightly changed risk metrics.
- Pure price/trend variants did not dominate the equal-weight composite in the final Backtrader comparison.
- ROE/P/E coverage depends on the frozen fundamental or BIST ratio file, so these factors can have shorter samples than momentum/trend.



Monte Carlo Significance

- The null model randomly selects stocks each month while preserving the number of positions.
- The p-value is the fraction of random portfolios with Sharpe at least as high as the strategy Sharpe.
- Saved p-value: 0.9045; this does not reject the random-strategy null at 5%.



Conclusion

- Best Backtrader run by final value: base with final value 3078262 TL.
- Public financial-statement fundamentals have limited history for some BIST stocks; results should be interpreted with that limitation.
- The final conclusion is cautious: the base Backtrader result is strong, but the Monte Carlo p-value and data limitations argue against real-money investment without better data and cost assumptions.